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Internet Security

Description:

This challenge lab puts together the conncepts that had been taught throughout the semester and more, and boy was this a challenge lab. Besides expanding upon what we had learned in the classroom we this lab also expands upon topics that were not covered in the classroom like VPN tunneling.

Tests:

Hosts connectivity:

Fire	Last Stat	Sourc	Destinati	Type	Color	Time(s	Perioc	Num	Edit	Delete
	Succe...	PC...	PC1C ...	ICMP		0.000	N	0	(edit)	
	Succe...	PC...	CorpC	ICMP		0.000	N	1	(edit)	
	Succe...	ISP...	ISP	ICMP		0.000	N	2	(edit)	
	Succe...	Cor...	CorpA	ICMP		0.000	N	3	(edit)	
	Succe...	PC1A	Corp-S...	ICMP		0.000	N	4	(edit)	
	Succe...	PC2A	PC1A	ICMP		0.000	N	5	(edit)	
	Succe...	PC2A	CorpA	ICMP		0.000	N	6	(edit)	
	Succe...	Cor...	CorpB-...	ICMP		0.000	N	7	(edit)	
	Failed	Cor...	PC2B ...	ICMP		0.000	N	8	(edit)	
	Succe...	PC3A	Corp-S...	ICMP		0.000	N	9	(edit)	
	Succe...	PC2A	Corp-S...	ICMP		0.000	N	10	(edit)	

Vlans:

Using the command sh vlan, I am able to demonstrate the vlan configurations that I have done for the different devices within the topology.

CorpA-SW:

```
CorpA-SW>enable
Password:
CorpA-SW#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
20   VLAN20                           active    Fa0/20
21   VLAN21                           active
22   VLAN22                           active
23   NetMgmt                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
21   enet  100021     1500  -      -      -        -    -        0      0
22   enet  100022     1500  -      -      -        -    -        0      0
23   enet  100023     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
```

CorpA-SW1:

```
*Mar 01, 04:55:47.5555: %SYS-5-CONFIG_I: Configured from console by console
CorpA-Sw1#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
20   VLAN20                           active
21   VLAN21                           active    Fa0/11
22   VLAN22                           active    Fa0/12
23   NetMgmt                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
21   enet  100021     1500  -      -      -        -    -        0      0
22   enet  100022     1500  -      -      -        -    -        0      0
23   enet  100023     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpA-Sw1#
```

CorpA-SW2:

```
CorpA-Sw2>enable
Password:
CorpA-Sw2#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
20   VLAN20                           active
21   VLAN21                           active    Fa0/11
22   VLAN22                           active    Fa0/12
23   NetMgmt                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
21   enet  100021     1500  -      -      -        -    -        0      0
22   enet  100022     1500  -      -      -        -    -        0      0
23   enet  100023     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpA-Sw2#
```

Corp-B Network

DMZ:

```
CorpB-DMZ
%SYS-5-CONFIG_I: Configured from console by console

CorpB-DMZ#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
23   NetMgmt                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
23   enet  100023     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpB-DMZ#
```

CorpB-Router:

```
CorpB
Physical  Config  CLI  Attributes
IOS Command Line Interface
CorpB-Router#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpB-Router#
```

CorpB-R1:

```
CorpB-R1>enable
Password:
CorpB-R1#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpB-R1#
```

CorpB-R2

```
CorpB-R2#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpB-R2#
```

CorpB-Sw:

```
CorpB-Sw#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
23   NetMgmt                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
23   enet  100023     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpB-Sw#
```

ISP:

```
ISP-Router#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
ISP-Router#
```

ISP-SW:

CorpC-Router

```
CorpC-Router#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpC-Router#
```

CorpC-Sw:

```
CorpC-Sw#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
23   NetMgmt                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
23   enet  100023     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
CorpC-Sw#
```

STP:

Using the command show spanning-tree I am able to display the spanning tree protocols on the CorpA-SW.

CorpA-Sw should be the STP root bridge for the VLANs within the Poughkeepsie

Location.

```
Password:
CorpA-SW#sh spanning-tree
VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    24596
             Address     0000.0C47.E0E6
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24596  (priority 24576 sys-id-ext 20)
             Address     0000.0C47.E0E6
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/11           Desg FWD 19        128.11   P2p
Fa0/12           Desg FWD 19        128.12   P2p
Fa0/20           Desg FWD 19        128.20   P2p

VLAN0021
  Spanning tree enabled protocol ieee
  Root ID    Priority    24597
             Address     0000.0C47.E0E6
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24597  (priority 24576 sys-id-ext 21)
             Address     0000.0C47.E0E6
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

```
VLAN0021
  Spanning tree enabled protocol ieee
  Root ID    Priority    24597
             Address     0000.0C47.E0E6
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24597  (priority 24576 sys-id-ext 21)
             Address     0000.0C47.E0E6
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/11           Desg FWD 19        128.11   P2p
Fa0/12           Desg FWD 19        128.12   P2p

VLAN0022
  Spanning tree enabled protocol ieee
  Root ID    Priority    24598
             Address     0000.0C47.E0E6
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24598  (priority 24576 sys-id-ext 22)
             Address     0000.0C47.E0E6
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/11           Desg FWD 19        128.11   P2p
Fa0/12           Desg FWD 19        128.12   P2p

VLAN0023
  Spanning tree enabled protocol ieee
  Root ID    Priority    24599
             Address     0000.0C47.E0E6
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24599  (priority 24576 sys-id-ext 23)
             Address     0000.0C47.E0E6
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/11           Desg FWD 19        128.11   P2p
Fa0/12           Desg FWD 19        128.12   P2p
```

CorpA-SW2:

```
Password:
CorpA-Sw2#sh spanning-tree
VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    24596
             Address     0000.0C47.E0E6
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     00D0.5843.A491
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------------
Fa0/1            Root FWD 19        128.1    P2p

VLAN0021
  Spanning tree enabled protocol ieee
  Root ID    Priority    24597
             Address     0000.0C47.E0E6
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32789  (priority 32768 sys-id-ext 21)
             Address     00D0.5843.A491
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/11           Desg FWD 19        128.11   P2p

VLAN0022
  Spanning tree enabled protocol ieee
  Root ID    Priority    24598
             Address     0000.0C47.E0E6
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/11           Desg FWD 19        128.11   P2p

VLAN0022
  Spanning tree enabled protocol ieee
  Root ID    Priority    24598
             Address     0000.0C47.E0E6
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32790  (priority 32768 sys-id-ext 22)
             Address     00D0.5843.A491
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/12           Desg FWD 19        128.12   P2p

VLAN0023
  Spanning tree enabled protocol ieee
  Root ID    Priority    24599
             Address     0000.0C47.E0E6
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32791  (priority 32768 sys-id-ext 23)
             Address     00D0.5843.A491
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------------
Fa0/1            Root FWD 19        128.1    P2p

CorpA-Sw2#
```

While this specific type of trunking does not need to be specified when configuring it on the devices in a network, at least for the types of switches I used in my topology, they still show up in th edisplay table when using th ecommand sh int trunk.

IEEE 802.1Q Trunking

CorpA-SW1

```
Press RETURN to get started!

CorpA-Sw1>
CorpA-Sw1>enable
Password:
CorpA-Sw1#sh int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      23

Port        Vlans allowed on trunk
Fa0/1       20-23

Port        Vlans allowed and active in management domain
Fa0/1       20,21,22,23

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       20,21,22,23

CorpA-Sw1#
```

CorpA-Sw2

```
CorpA-Sw2>enable
Password:
CorpA-Sw2#sh int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      23

Port        Vlans allowed on trunk
Fa0/1       20-23

Port        Vlans allowed and active in management domain
Fa0/1       20,21,22,23

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       20,21,22,23

CorpA-Sw2#
```

CorpA-SW

Physical Config CLI Attributes

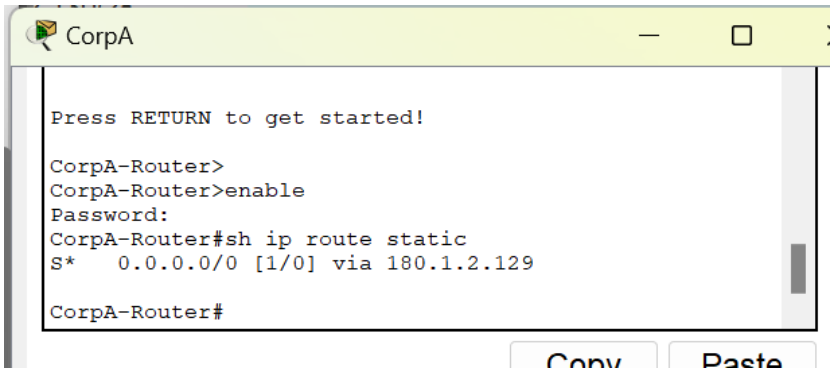
IOS Command Line Interface

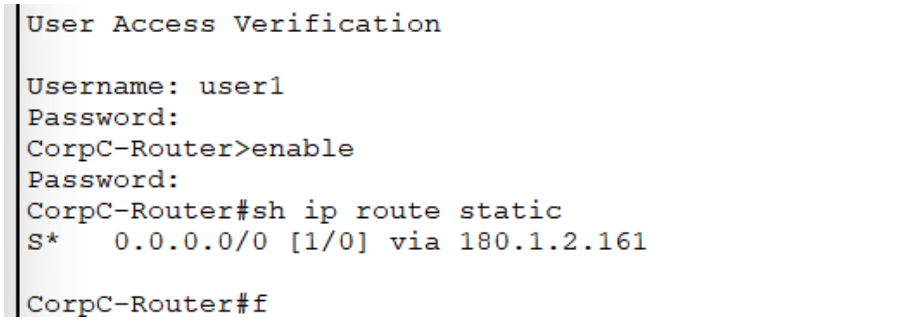
```
Press RETURN to get started!

CorpA-SW>enable
Password:
CorpA-SW#sh int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      23
Fa0/11      on           802.1q         trunking      23
Fa0/12      on           802.1q         trunking      23

Port        Vlans allowed on trunk
Fa0/1       20-23
Fa0/11      20-23
Fa0/12      20-23

Port        Vlans allowed and active in management domain
Fa0/1       20,21,22,23
Fa0/11      20,21,22,23
Fa0/12      20,21,22,23

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       20,21,22,23
Fa0/11      20,21,22,23
Fa0/12      20,21,22,23

CorpA-SW#
```

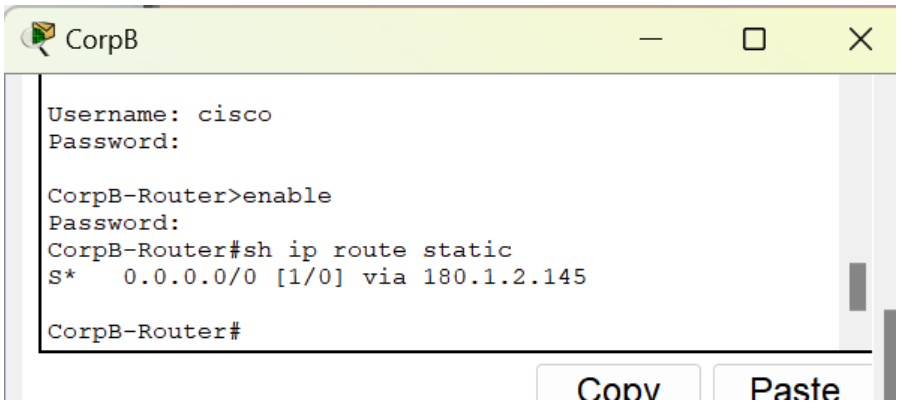
Default Static Routing

Using th ecommand sh ip route static, or even just sh ip route, I am able to display the static routes configured, and when specifying the command to only display the static route, it makes it easier to read.

✓ The Edge Routers should have a default static route to the ISP router.
✓ You should determine which routers are Edge Routers.







Dynamic Routing

Like with the sh ip route static, the sh ip route shows the static route, but it also shows the dynamically configured routes, the CorpB devices. To be even more specific with displaying OSPF, I used the command sh ip ospf.

Corp B network

```
CorpB-Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 180.1.2.145 to network 0.0.0.0

     172.28.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.28.30.0/24 is directly connected, GigabitEthernet0/0
L       172.28.30.1/32 is directly connected, GigabitEthernet0/0
C       172.28.31.32/30 is directly connected, Serial0/0/0
L       172.28.31.33/32 is directly connected, Serial0/0/0
O       172.28.31.36/30 [110/128] via 172.28.31.34, 00:35:49, Serial0/0/0
                        [110/128] via 172.28.31.41, 00:35:49, Serial0/1/0
C       172.28.31.40/30 is directly connected, Serial0/1/0
L       172.28.31.42/32 is directly connected, Serial0/1/0
C       172.28.99.100/32 is directly connected, Loopback0
     180.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       180.1.2.144/28 is directly connected, Serial0/0/1
L       180.1.2.146/32 is directly connected, Serial0/0/1
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/1
L       192.168.30.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 180.1.2.145

CorpB-Router#
```

Copy Paste

```
Password:
CorpB-R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.28.31.42 to network 0.0.0.0

     172.28.0.0/16 is variably subnetted, 7 subnets, 3 masks
O       172.28.30.0/24 [110/65] via 172.28.31.42, 00:39:26, Serial0/1/0
O       172.28.31.32/30 [110/128] via 172.28.31.42, 00:39:26, Serial0/1/0
                        [110/128] via 172.28.31.37, 00:39:26, Serial0/0/1
C       172.28.31.36/30 is directly connected, Serial0/0/1
L       172.28.31.37/32 is directly connected, Serial0/0/1
C       172.28.31.40/30 is directly connected, Serial0/1/0
L       172.28.31.41/32 is directly connected, Serial0/1/0
C       172.28.99.102/32 is directly connected, Loopback0
O    192.168.30.0/24 [110/65] via 172.28.31.42, 00:39:26, Serial0/1/0
O*E2 0.0.0.0/0 [110/1] via 172.28.31.42, 00:39:26, Serial0/1/0

CorpB-R2#
```

```
CorpB-R1>enable
Password:
CorpB-R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.28.31.33 to network 0.0.0.0

     172.28.0.0/16 is variably subnetted, 7 subnets, 3 masks
O       172.28.30.0/24 [110/65] via 172.28.31.33, 00:40:20, Serial0/0/0
C       172.28.31.32/30 is directly connected, Serial0/0/0
L       172.28.31.34/32 is directly connected, Serial0/0/0
C       172.28.31.36/30 is directly connected, Serial0/0/1
L       172.28.31.37/32 is directly connected, Serial0/0/1
O       172.28.31.40/30 [110/128] via 172.28.31.37, 00:40:20, Serial0/0/1
                        [110/128] via 172.28.31.33, 00:40:20, Serial0/0/0
C       172.28.99.101/32 is directly connected, Loopback0
O    192.168.30.0/24 [110/65] via 172.28.31.33, 00:40:20, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 172.28.31.33, 00:40:20, Serial0/0/0

CorpB-R1#
```

```
ISP-Router>enable
Password:
ISP-Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     180.1.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       180.1.2.80/29 is directly connected, GigabitEthernet0/0
L       180.1.2.81/32 is directly connected, GigabitEthernet0/0
C       180.1.2.128/28 is directly connected, Serial0/0/0
L       180.1.2.129/32 is directly connected, Serial0/0/0
C       180.1.2.160/28 is directly connected, Serial0/1/0
L       180.1.2.161/32 is directly connected, Serial0/1/0

ISP-Router#
```

```
CorpB-R1>enable
Password:
CorpB-R1#sh ip ospf
 Routing Process "ospf 1" with ID 172.28.99.101
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 1. Checksum Sum 0x000235
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 2
        Area has message digest authentication
        SPF algorithm executed 28 times
        Area ranges are
        Number of LSA 3. Checksum Sum 0x01c78d
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

CorpB-R1#
```

```
CorpB-R2>enable
Password:
CorpB-R2#sh ip ospf
 Routing Process "ospf 1" with ID 172.28.99.102
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 1. Checksum Sum 0x000235
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 2
        Area has message digest authentication
        SPF algorithm executed 30 times
        Area ranges are
        Number of LSA 3. Checksum Sum 0x01c78d
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

CorpB-R2#
```

Default Route Injection

The command sh ip ospf database displays the default route injection, and the ospf details.

```
CorpB-R1## Command
           ^
% Invalid input detected at '^' marker.

CorpB-R1#show ip ospf database
            OSPF Router with ID (172.28.99.101) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
172.28.99.101   172.28.99.101   574         0x80000013 0x00d8d1 4
172.28.99.100   172.28.99.100   1362        0x80000013 0x00b35d 6
172.28.99.102   172.28.99.102   577         0x80000013 0x003b5f 4

                Type-5 AS External Link States
Link ID         ADV Router      Age         Seq#       Checksum Tag
0.0.0.0         172.28.99.100   580         0x8000000c 0x000235 1
CorpB-R1#
```

```
CorpB-R2#sh ip ospf database
            OSPF Router with ID (172.28.99.102) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
172.28.99.102   172.28.99.102   617         0x80000013 0x003b5f 4
172.28.99.100   172.28.99.100   1402        0x80000013 0x00b35d 6
172.28.99.101   172.28.99.101   615         0x80000013 0x00d8d1 4

                Type-5 AS External Link States
Link ID         ADV Router      Age         Seq#       Checksum Tag
0.0.0.0         172.28.99.100   620         0x8000000c 0x000235 1
CorpB-R2#
```

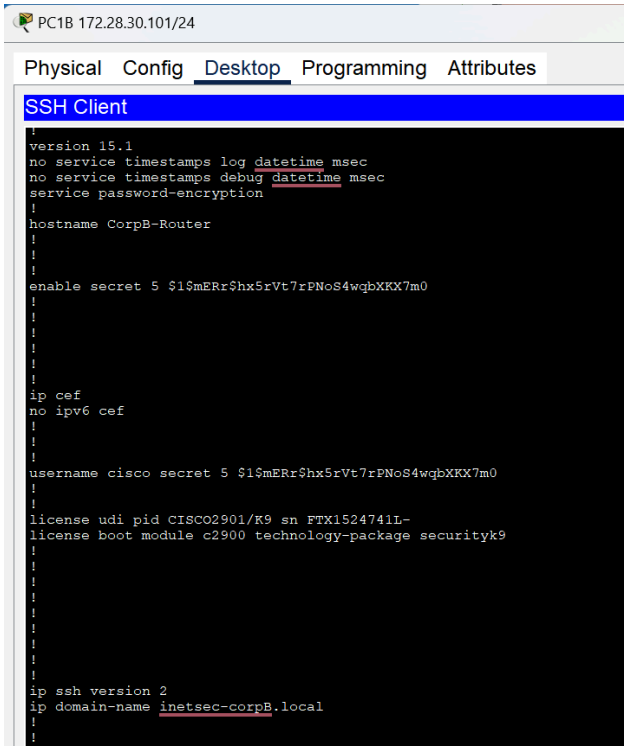
DMZ Access Control Lists

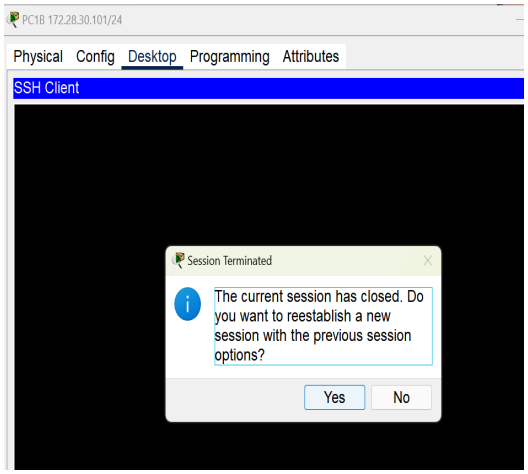
```
CorpB-DMZ#
CorpB-DMZ#sh access-list
Extended IP access list SSH-Access
    10 permit tcp host 192.168.30.200 any eq 22
    20 deny tcp any any eq 22
    30 permit ip any any

CorpB-DMZ#
```

SSH

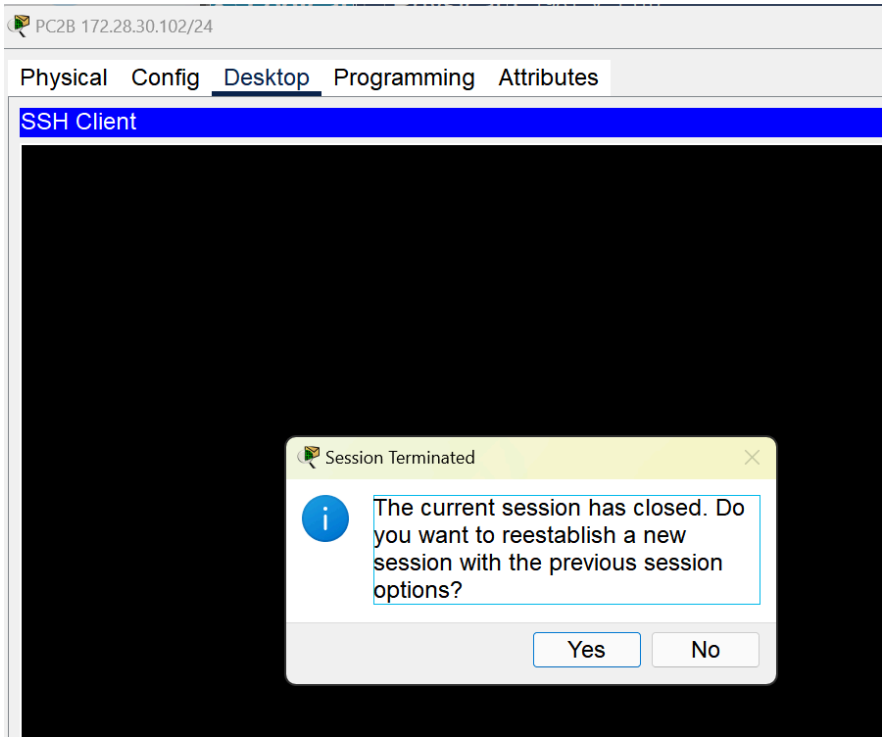
✓ Enable SSH on all devices that support it and ensure only one PC per network is allowed to access all devices via SSH.

✓ Telnet should not be allowed.

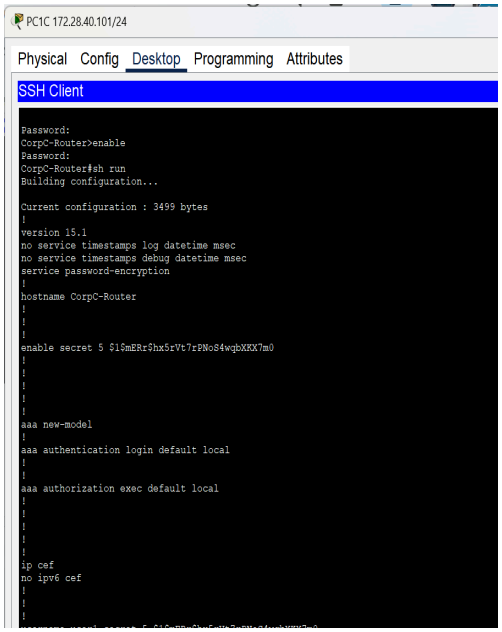


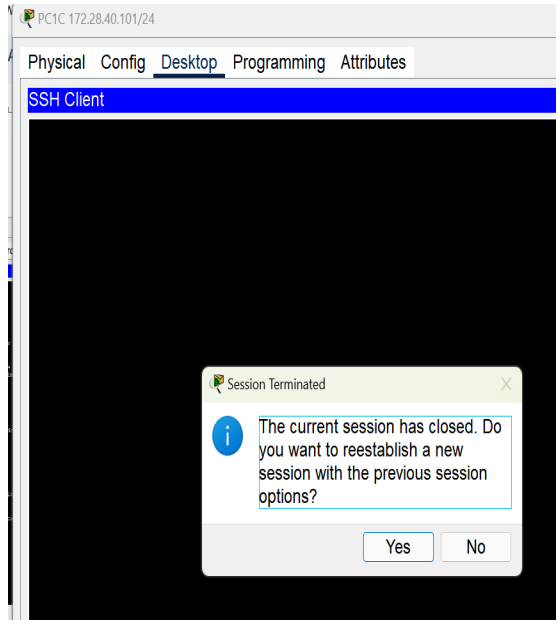


I was able to ssh into router CorpB from PC1B but I was not able to telnet into the same router from the same PC.



I was not able to ssh or telnet into CorpB router from PC2B.





I was able to ssh into the CorpC router from PC1C, but I was not able to telnet into the same router. I was not able to ssh or telnet into the CorpC router from PC2C.

DHCP

✓ Configure the DHCP server for the appropriate networks according to the network topology.

✓ Ensure the relay agent is in place, if required.

✓ The IP Address range should begin with the IP Address specified on the specific device, such as a PC.

✓ The gateway should be set to the first available address within the appropriate range.

✓ Make sure a DNS server and DNS suffix is specified within every DHCP scope configured.

CorpA-Router

```
CorpA-Router#
CorpA-Router#sh ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
172.28.21.101    0001.9647.3E11           --
Automatic
172.28.21.102    00D0.BA83.02C7           --
Automatic
172.28.22.101    0002.167E.1E9E           --
Automatic
172.28.22.102    0040.0B2C.8424           --
Automatic
CorpA-Router#
```

DNS

CorpA

Physical Config CLI Attributes

IOS Command L

```
version 15.1
service timestamps log datetime msec
service timestamps debug datetime msec
service password-encryption
!
hostname CorpA-Router
!
!
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
ip dhcp excluded-address 172.28.21.1 172.28.21.
ip dhcp excluded-address 172.28.22.1 172.28.22.
!
ip dhcp pool VLAN21-POOL
 network 172.28.21.0 255.255.255.0
 default-router 172.28.21.1
 dns-server 172.28.20.21
 domain-name inetsec-corpA.local
ip dhcp pool VLAN22-POOL
 network 172.28.22.0 255.255.255.0
 default-router 172.28.22.1
 dns-server 172.28.20.21
 domain-name inetsec-corpA.local
!
```

PAT

✓ Port Address Translation should be configured to use the interface ip address on the CorpA router.

i. All devices configured with RFC1918 addressing should have the ability to gain connectivity to the Internet.

Web Browser X

< > URL http://180.1.2.85 Go Stop

Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:
A small page
Copyrights
Image page
Image

```
CorpA-Router#sh ip nat tran
Pro  Inside global     Inside local       Outside local      Outside global
tcp 180.1.2.130:1025   172.28.21.101:1025 180.1.2.85:80      180.1.2.85:80

CorpA-Router#
```

```
Username: user1
Password:
CorpC-Router>enable
Password:
CorpC-Router#sh ip nat tran
Pro  Inside global     Inside local       Outside local      Outside
global
tcp 180.1.2.162:1026   172.28.40.102:1026 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.162:1027   172.28.40.102:1027 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.162:1028   172.28.40.102:1028 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.162:1030   172.28.40.101:1030 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.162:1035   172.28.40.101:1035 180.1.2.80:80      180.1.2.80:80
tcp 180.1.2.162:1036   172.28.40.101:1036 180.1.2.80:80      180.1.2.80:80
tcp 180.1.2.162:1037   172.28.40.101:1037 180.1.2.80:80      180.1.2.80:80
tcp 180.1.2.162:1038   172.28.40.101:1038 180.1.2.80:80      180.1.2.80:80
tcp 180.1.2.162:1042   172.28.40.101:1042 180.1.2.85:80      180.1.2.85:80

CorpC-Router#
```

```
CorpB-Router>enable
Password:
CorpB-Router#sh ip nat tran
Pro  Inside global     Inside local       Outside local      Outside global
---  180.1.2.157       192.168.30.100     ---                ---
---  180.1.2.158       192.168.30.100     ---                ---
tcp 180.1.2.147:1024   172.28.30.101:1029 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1025   172.28.30.101:1030 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1027   172.28.30.102:1027 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1028   172.28.30.102:1028 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1029   172.28.30.102:1029 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1030   172.28.30.102:1030 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1031   172.28.30.102:1031 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1032   172.28.30.102:1032 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1033   172.28.30.102:1033 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1034   172.28.30.102:1034 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1035   172.28.30.102:1035 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1036   172.28.30.102:1036 180.1.2.85:80      180.1.2.85:80
tcp 180.1.2.147:1037   172.28.30.102:1037 180.1.2.85:80      180.1.2.85:80

CorpB-Router#
```

HTTP Server Static NAT

✓ Static Network Address Translations should be configured so that one public IP Address is bound to the CorpB-HTTP-Server.

✓ All traffic destined for the CorpB-HTTP-Server via HTTP should be permitted from either CorpB's LAN or the Internet.

✓ All other traffic to this server should be denied.

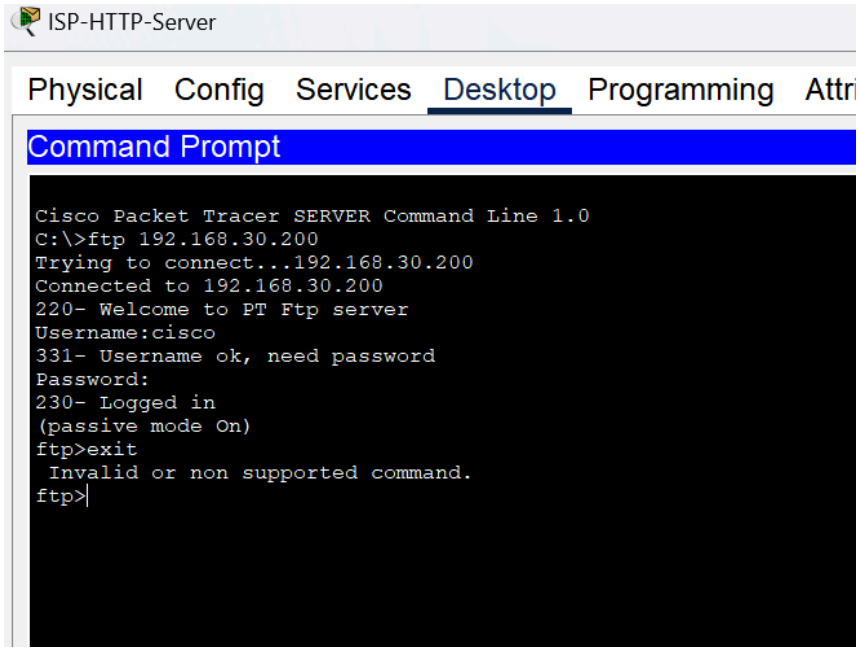
```
CorpB-Router#sh ip access-list
Extended IP access list SSH-Access
    10 permit tcp host 172.28.30.101 any eq 22 (4 match(es))
    20 deny tcp any any eq 22
Standard IP access list CORPB-NAT-ACL
    10 permit 172.28.30.0 0.0.0.255 (72 match(es))
    20 permit 192.168.30.0 0.0.0.255
    30 permit 172.28.31.32 0.0.0.3
    40 permit 172.28.31.36 0.0.0.3
    50 permit 172.28.31.40 0.0.0.3
    60 deny any (48 match(es))
Extended IP access list SERVER-ACCESS
    10 permit tcp 172.28.30.0 0.0.0.255 host 192.168.30.100 eq www
    20 permit tcp 192.168.30.0 0.0.0.255 host 192.168.30.100 eq www
    30 permit tcp 172.28.31.32 0.0.0.3 host 192.168.30.100 eq www
    40 permit tcp 172.28.31.36 0.0.0.3 host 192.168.30.100 eq www
    50 permit tcp 172.28.31.40 0.0.0.3 host 192.168.30.100 eq www
    60 permit tcp 180.1.2.80 0.0.0.7 host 192.168.30.100 eq www
    70 permit tcp 172.28.30.0 0.0.0.255 host 192.168.30.200 eq ftp
    80 permit tcp 192.168.30.0 0.0.0.255 host 192.168.30.200 eq ftp
    90 permit tcp 172.28.31.32 0.0.0.3 host 192.168.30.200 eq ftp
    100 permit tcp 172.28.31.36 0.0.0.3 host 192.168.30.200 eq ftp
    110 permit tcp 172.28.31.40 0.0.0.3 host 192.168.30.200 eq ftp
    120 permit tcp 180.1.2.80 0.0.0.7 host 192.168.30.200 eq ftp
    130 permit tcp 172.28.30.0 0.0.0.255 host 192.168.30.200 eq 20
    140 permit tcp 192.168.30.0 0.0.0.255 host 192.168.30.200 eq 20
    150 permit tcp 172.28.31.32 0.0.0.3 host 192.168.30.200 eq 20
    160 permit tcp 172.28.31.36 0.0.0.3 host 192.168.30.200 eq 20
    170 permit tcp 172.28.31.40 0.0.0.3 host 192.168.30.200 eq 20
    180 permit tcp 180.1.2.80 0.0.0.7 host 192.168.30.200 eq 20
    190 deny ip any any (24 match(es))
Extended IP access list RESPONSE-TRAFFIC
    10 permit tcp any any established
    20 deny ip any any (17 match(es))

CorpB-Router#
```

```
CorpA-Router#
*Mar 02, 10:25:50.2525: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0,
changed state to down
*Mar 02, 10:25:50.2525: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0,
changed state to up
CorpA-Router#sh ip access-list
Extended IP access list SSH-Access
    10 permit tcp host 172.28.20.21 any eq 22
    20 deny tcp any any eq 22 (5 match(es))
    30 deny ip any any
Extended IP access list INBOUND
    10 permit ip 172.28.40.0 0.0.0.255 172.28.20.0 0.0.0.255
    20 permit ip 172.28.40.0 0.0.0.255 172.28.21.0 0.0.0.255
    30 permit ip 172.28.40.0 0.0.0.255 172.28.22.0 0.0.0.255
    40 permit ip 172.28.40.0 0.0.0.255 172.28.23.0 0.0.0.255
    50 permit ip host 180.1.2.162 host 180.1.2.130 (2709 match(es))
    60 permit icmp host 180.1.2.162 host 180.1.2.130
    70 permit udp host 180.1.2.162 host 180.1.2.130 eq isakmp
    80 permit esp host 180.1.2.162 host 180.1.2.130
    90 permit ahp host 180.1.2.162 host 180.1.2.130
    100 permit ip host 180.1.2.129 host 180.1.2.130 (4 match(es))
    110 permit tcp 180.1.2.80 0.0.0.7 any established
    120 deny ip any any (7 match(es))
Extended IP access list 100
    10 permit ip 172.28.20.0 0.0.0.255 172.28.40.0 0.0.0.255 (1 match(es))
    20 permit ip 172.28.21.0 0.0.0.255 172.28.40.0 0.0.0.255
    30 permit ip 172.28.22.0 0.0.0.255 172.28.40.0 0.0.0.255
    40 permit ip 172.28.23.0 0.0.0.255 172.28.40.0 0.0.0.255
Extended IP access list CORPA-PAT
    10 deny ip 172.28.20.0 0.0.0.255 172.28.40.0 0.0.0.255 (2 match(es))
    20 deny ip 172.28.21.0 0.0.0.255 172.28.40.0 0.0.0.255
    30 deny ip 172.28.22.0 0.0.0.255 172.28.40.0 0.0.0.255
    40 deny ip 172.28.23.0 0.0.0.255 172.28.40.0 0.0.0.255
    50 permit ip 172.28.20.0 0.0.0.255 any (346 match(es))
    60 permit ip 172.28.21.0 0.0.0.255 any (16 match(es))
    70 permit ip 172.28.22.0 0.0.0.255 any
    80 permit ip 172.28.23.0 0.0.0.255 any

CorpA-Router#
```

```
CorpC-Router#sh ip access-lists
Extended IP access list CORPC-NAT
    10 deny ip 172.28.40.0 0.0.0.255 172.28.20.0 0.0.0.255 (2 match(es))
    20 deny ip 172.28.40.0 0.0.0.255 172.28.21.0 0.0.0.255 (8 match(es))
    30 deny ip 172.28.40.0 0.0.0.255 172.28.22.0 0.0.0.255
    40 deny ip 172.28.40.0 0.0.0.255 172.28.23.0 0.0.0.255
    50 permit ip 172.28.40.0 0.0.0.255 any (50 match(es))
Extended IP access list INBOUND
    10 permit ip 172.28.20.0 0.0.0.255 172.28.40.0 0.0.0.255
    20 permit ip 172.28.21.0 0.0.0.255 172.28.40.0 0.0.0.255
    30 permit ip 172.28.22.0 0.0.0.255 172.28.40.0 0.0.0.255
    40 permit ip 172.28.23.0 0.0.0.255 172.28.40.0 0.0.0.255
    50 permit ip host 180.1.2.130 host 180.1.2.162 (2700 match(es))
    60 permit ip host 180.1.2.161 host 180.1.2.162 (2 match(es))
    70 permit ip host 180.1.2.146 host 180.1.2.160
    80 permit tcp 180.1.2.80 0.0.0.7 any established (4 match(es))
    90 deny ip any any (8 match(es))
Extended IP access list SSH-Access
    10 permit tcp host 172.28.40.101 any eq 22 (8 match(es))
    20 deny tcp any any eq 22
Extended IP access list 100
    10 permit ip 172.28.40.0 0.0.0.255 172.28.20.0 0.0.0.255 (1 match(es))
    20 permit ip 172.28.40.0 0.0.0.255 172.28.21.0 0.0.0.255 (1 match(es))
    30 permit ip 172.28.40.0 0.0.0.255 172.28.22.0 0.0.0.255
    40 permit ip 172.28.40.0 0.0.0.255 172.28.23.0 0.0.0.255

CorpC-Router#
```

FTP Server Static NAT

I am able to FTP into my corpB-FTP-server form my ISP server.



Zone-Based Firewall

✓ The CorpB router should be configured as a Zone-Based Firewall.

✓ All traffic from the Internet to the CorpB LAN should be denied by default.

✓ If the traffic is initiated from within the LAN and destined for the Internet, the return traffic should be permitted to establish a TCP session and send/receive data.

✓ Once the TCP session is completed and torn down, the return traffic should be denied (if initiated).

```
CorpB-Router>enable
Password:
CorpB-Router#sh zone sec
zone self
  Description: System defined zone


zone PRIVATE
  Member Interfaces:
    GigabitEthernet0/0
    Serial0/0/0
    Serial0/1/0


zone PUBLIC
  Member Interfaces:
    Serial0/0/1


CorpB-Router#sh class-map
 Class Map match-any class-default (id 0)
   Match any
CorpB-Router#
```

Local AAA

CorpC

Physical Config CLI Attributes

IOS Command Line Interface

```
                                   ^
% Invalid input detected at '^' marker.

CorpC-Router#show aaa user
% Incomplete command.
CorpC-Router#show aaa user ?
  <1-4294967295>  Unique Id for a single user
  all             Show users active in AAA subsystem
CorpC-Router#show aaa user all
----------------------------------
Unique id 22 is currently in use.
Accounting:
log=0x18001
Events recorded :
        CALL START
        INTERIM START
        INTERIM STOP
update method(s) :
        NONE
update interval = 0
Outstanding Stop Records : 0
Dynamic attribute list:
        47B2AC48 0 00000001 connect-progress(52) 4 Term Serv Auth
        47B2AC58 0 00000001 pre-session-time(268) 4 814(32E)
        47B2AC68 0 00000001 elapsed_time(338) 4 0(0)
        47B2AC78 0 00000001 pre-bytes-in(264) 4 0(0)
        47B2AC88 0 00000001 pre-bytes-out(265) 4 0(0)
        47B2AC98 0 00000001 pre-paks-in(266) 4 0(0)
        47B2ACA8 0 00000001 pre-paks-out(267) 4 0(0)
        No data for type EXEC
        No data for type CONN
        NET: Username=(n/a)
                Session Id=0000016 Unique Id=0000017
                Start Sent=0 Stop Only=N
                stop_has_been_sent=N
                Method List=0
                Attribute list:
        47B2AC48 0 00000001 session-id(336) 4 23(17)
No data for type CMD
No data for type SYSTEM
No data for type RM CALL
No data for type RM VPDN
No data for type AUTH PROXY
No data for type 8
No data for type CALL
```

```
CorpC-Router#sh aaa sessions
Total sessions since last reload: 7
Session Id:23
        Unique Id:22
        User Name:user1
        IP Address:0.0.0.0
        Idle Time: 0
        CT Call Handle: 0
CorpC-Router#sh aaa user
% Incomplete command.
CorpC-Router#sh aaa users
```

18. Server-based AAA

✓ The Corp-Server should be configured as a AAA server using RADIUS for device authentication.

20. NTP

✓ The CorpA router should utilize the Corp-Server as an NTP server.

✓ Ensure that timestamps are enabled and being used for all supported devices on the CorpA network.

21. InterVLAN Routing

✓ CorpA should have interVLAN routing configured so VLAN23 is the administrative VLAN and named “NetMgmt”.

```
Device Name: CorpA
Device Model: 2901
Hostname: CorpA-Router

Port                    Link   VLAN   IP Address         IPv6 Address
GigabitEthernet0/0      Up     --     <not set>          <not set>
GigabitEthernet0/0.20   Up     --     172.28.20.1/24     <not set>
GigabitEthernet0/0.21   Up     --     172.28.21.1/24     <not set>
GigabitEthernet0/0.22   Up     --     172.28.22.1/24     <not set>
GigabitEthernet0/0.23   Up     --     172.28.23.1/24     <not set>
GigabitEthernet0/1      Down   --     <not set>          <not set>
Serial0/0/0             Up     --     180.1.2.130/28     <not set>
Serial0/0/1             Down   --     <not set>          <not set>
Serial0/1/0             Down   --     <not set>          <not set>
Serial0/1/1             Down   --     <not set>          <not set>
Vlan1                   Down   1      <not set>          <not set>
```

22. IPSec VPN

✓ CorpA and CorpC should establish a site-to-site VPN tunnel so that all CorpA PCs and Servers should be able to communicate with CorpC PCs without being translated (ie: using NAT or PAT).

23. VTP

```
CorpA-Router>enable
Password:
CorpA-Router#sh vtp
% Incomplete command.
CorpA-Router#sh vtp ?
  counters  VTP statistics
  status    VTP domain status
CorpA-Router#sh vtp status
VTP Version                     : 1
Configuration Revision          : 11
Maximum VLANs supported locally : 36
Number of existing VLANs        : 9
VTP Operating Mode              : Server
VTP Domain Name                 :
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0x1E 0xD5 0xC6 0xE2 0xC3 0xFE 0x1D 0x56
Configuration last modified by 0.0.0.0 at 3-1-93 00:48:19
Local updater ID is 0.0.0.0 (no valid interface found)
CorpA-Router#
```

Copy Pa

Corp Switch

```
Username: user2
Password:
CorpA-SW>enable
Password:
CorpA-SW#sh vtp ?
  counters  VTP statistics
  password  VTP password
  status    VTP domain status
CorpA-SW#sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : CorpA
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 00E0.F91C.7200
Configuration last modified by 172.28.23.10 at 3-1-93 01:00:51
Local updater ID is 172.28.23.10 on interface Vl23 (lowest numbered VLAN interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 9
Configuration Revision            : 12
MD5 digest                        : 0x1C 0xBD 0x9A 0xFF 0xFF 0xCD 0x4D 0xBB
                                    0xD9 0xE1 0x4B 0x0B 0x69 0x8E 0x41 0xE3
CorpA-SW#sh vtp counters
VTP statistics:
Summary advertisements received    : 275
Subset advertisements received     : 1
Request advertisements received    : 0
Summary advertisements transmitted : 156
Subset advertisements transmitted  : 13
Request advertisements transmitted : 0
Number of config revision errors   : 0
Number of config digest errors     : 0
Number of V1 summary errors        : 0


VTP pruning statistics:

Trunk            Join Transmitted Join Received    Summary advts received from
                                                   non-pruning-capable device
---------------- ---------------- ---------------- ---------------------------
CorpA-SW#Username:
CorpA-SW#sh vtp password
VTP Password: cisco
CorpA-SW#
```

25. Other Protocols

✓ All routers should be locked down as much as possible.

✓ Only enable protocols that need to be enabled.

✓ All switches should be locked down as much as possible.

✓ Any interfaces not in use should be disabled.

Device Name: CorpC-SW
Custom Device Model: 2960 IOS15
Hostname: CorpC-Sw

Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	1	--	0001.6304.8D01
FastEthernet0/2	Down	1	--	0001.6304.8D02
FastEthernet0/3	Down	1	--	0001.6304.8D03
FastEthernet0/4	Down	1	--	0001.6304.8D04
FastEthernet0/5	Down	1	--	0001.6304.8D05
FastEthernet0/6	Down	1	--	0001.6304.8D06
FastEthernet0/7	Down	1	--	0001.6304.8D07
FastEthernet0/8	Down	1	--	0001.6304.8D08
FastEthernet0/9	Down	1	--	0001.6304.8D09
FastEthernet0/10	Down	1	--	0001.6304.8D0A
FastEthernet0/11	Up	1	--	0001.6304.8D0B
FastEthernet0/12	Up	1	--	0001.6304.8D0C
FastEthernet0/13	Down	1	--	0001.6304.8D0D
FastEthernet0/14	Down	1	--	0001.6304.8D0E
FastEthernet0/15	Down	1	--	0001.6304.8D0F
FastEthernet0/16	Down	1	--	0001.6304.8D10
FastEthernet0/17	Down	1	--	0001.6304.8D11
FastEthernet0/18	Down	1	--	0001.6304.8D12
FastEthernet0/19	Down	1	--	0001.6304.8D13
FastEthernet0/20	Down	1	--	0001.6304.8D14
FastEthernet0/21	Down	1	--	0001.6304.8D15
FastEthernet0/22	Down	1	--	0001.6304.8D16
FastEthernet0/23	Down	1	--	0001.6304.8D17
FastEthernet0/24	Down	1	--	0001.6304.8D18
GigabitEthernet0/1	Down	1	--	0001.6304.8D19
GigabitEthernet0/2	Down	1	--	0001.6304.8D1A
Vlan1	Down	1	<not set>	0005.5E1D.9796
Vlan23	Up	23	172.28.40.10/24	0005.5E1D.9701

Device Name: CorpA
Device Model: 2901
Hostname: CorpA-Router

Port	Link	VLAN	IP Address
GigabitEthernet0/0	Up	--	<not set>
GigabitEthernet0/0.20	Up	--	172.28.20.1/24
GigabitEthernet0/0.21	Up	--	172.28.21.1/24
GigabitEthernet0/0.22	Up	--	172.28.22.1/24
GigabitEthernet0/0.23	Up	--	172.28.23.1/24
GigabitEthernet0/1	Down	--	<not set>
Serial0/0/0	Up	--	180.1.2.130/28
Serial0/0/1	Down	--	<not set>
Serial0/1/0	Down	--	<not set>
Serial0/1/1	Down	--	<not set>
Vlan1	Down	1	<not set>

Physical Location: Intercity > Home City > Corporate

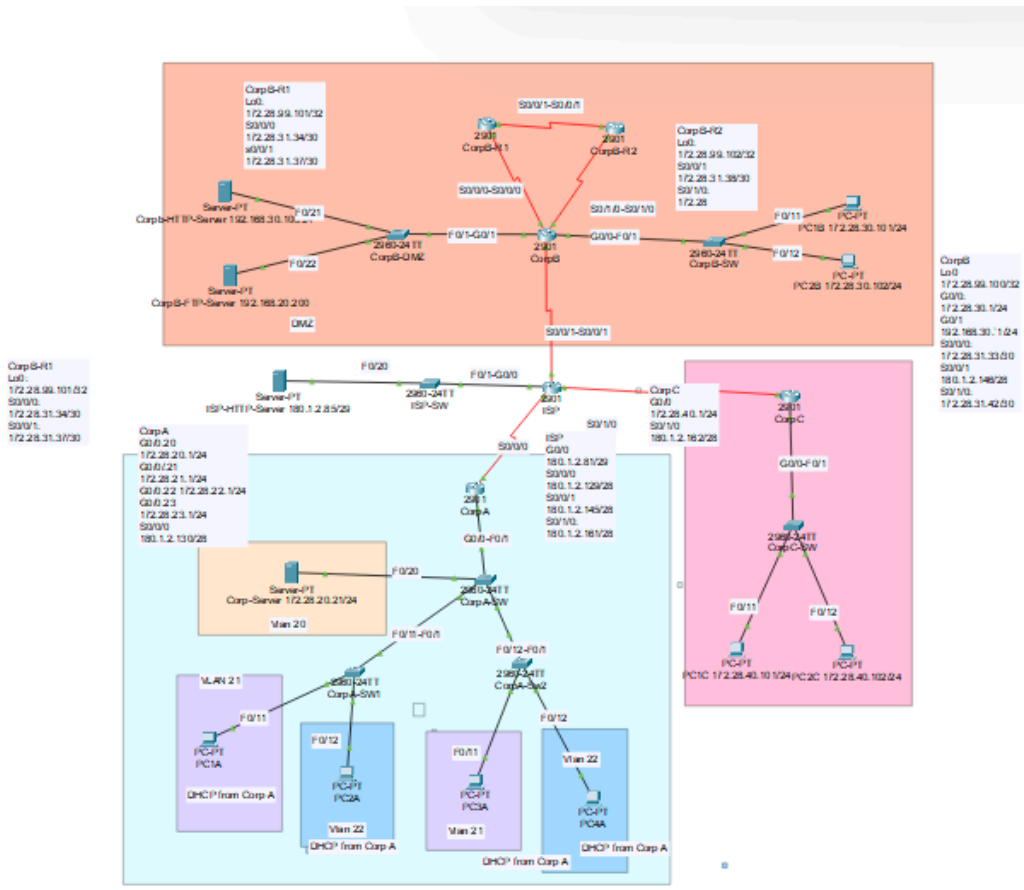
Device Name: CorpA-SW1
Custom Device Model: 2960 IOS15
Hostname: CorpA-Sw1

Port	Link	VLAN	IP Address	M
FastEthernet0/1	Up	--	--	0
FastEthernet0/2	Down	1	--	0
FastEthernet0/3	Down	1	--	0
FastEthernet0/4	Down	1	--	0
FastEthernet0/5	Down	1	--	0
FastEthernet0/6	Down	1	--	0
FastEthernet0/7	Down	1	--	0
FastEthernet0/8	Down	1	--	0
FastEthernet0/9	Down	1	--	0
FastEthernet0/10	Down	1	--	0
FastEthernet0/11	Up	21	--	0
FastEthernet0/12	Up	22	--	0
FastEthernet0/13	Down	999	--	0
FastEthernet0/14	Down	999	--	0
FastEthernet0/15	Down	999	--	0
FastEthernet0/16	Down	999	--	0
FastEthernet0/17	Down	999	--	0
FastEthernet0/18	Down	999	--	0
FastEthernet0/19	Down	999	--	0
FastEthernet0/20	Down	999	--	0
FastEthernet0/21	Down	999	--	0
FastEthernet0/22	Down	999	--	0
FastEthernet0/23	Down	999	--	0
FastEthernet0/24	Down	999	--	0
GigabitEthernet0/1	Down	999	--	0
GigabitEthernet0/2	Down	999	--	0
Vlan1	Down	1	<not set>	0
Vlan21	Up	21	<not set>	0
Vlan22	Up	22	<not set>	0
Vlan23	Up	23	172.28.23.11/24	0
Vlan999	Down	999	<not set>	0

Physical Location: Intercity > Home City > Corporate Of

```
Device Name: ISP
Device Model: 2901
Hostname: ISP-Router

Port                 Link   VLAN   IP Address         IPv6 Address
GigabitEthernet0/0   Up     --     180.1.2.81/29      <not set>
GigabitEthernet0/1   Down   --     <not set>          <not set>
Serial0/0/0          Up     --     180.1.2.129/28     <not set>
Serial0/0/1          Up     --     <not set>          <not set>
Serial0/1/0          Up     --     180.1.2.161/28     <not set>
Serial0/1/1          Down   --     <not set>          <not set>
Vlan1                Down   1      <not set>          <not set>

Physical Location: Intercity > Home City > Corporate Office > Main
```

Topology:



Conclusion:

This lab takes everything that has been taught and learned throughout the semester and compiled it into one lab, while not everything in this lab was taught in class, the skills that were taught in class gave me the stepping stone into learning how to perform these new configurations.